

# Unitree Week 7

Presentation by Mario Ortiz

# This week

- Decided to look into Isaac (Isaac Gym Preview 4 and IsaacLab+IsaacSim) as there are some examples of using reinforcement learning to simulate the g1 unitree robot walking.
- Some Gazebo research

# Gazebo Facts

- Gazebo is an open-source 3D robotics simulator which allows for accurate and efficient simulation of robots.
- Gazebo has:
  - Various physics Engines (ODE, Bullet, DART, Simbody)
  - Sensor Simulation: can generate realistic data from various sensors (LiDAR for example)
  - ROS Integration: Gazebo acts as the simulation engine for Robot Operating System (ROS)
  - Supports various plugins for designing customized robots, sensors, and environment control.

# Gazebo Facts

- Recent information: Gazebo Classic (Versions 1-11) are deprecated.
  - Gazebo development has moved to Gazebo Ignition (or modular Gazebo)
- Authors: Nate Koenig and Andrew Howard (2002)
- Why use Gazebo?
  - Doesn't require an NVIDIA GPU
  - Seamless integration with ROS

# Isaac Gym and Isaac Lab overview

- Isaac Gym Preview 4: a standalone reinforcement learning framework
  - The sole purpose is fast, rigid-body physics.
  - Deprecated, so no more updates/improvements to the software (Isaac Lab is what is ongoing/updating)
- Isaac Lab: Modern open-source framework built on Isaac Sim.
  - Supports more complex, high-fidelity environments, sensor simulation, as Robot Operating System (ROS).
  - Successor of Isaac Gym Preview 4

Both of these contain examples of using RL to train the g1 to walk. The goal is to learn how to implement/use these examples, and repurpose them into the r1 robot (how can we utilize the methods to train the g1 to train the r1 in the future?)

# Using Isaac Gym Preview 4

- Install Ubuntu 20.04 (will use windows subsystem linux).
  - My laptop has an Nvidia RTX 3080, so I will use windows subsystem linux to use Ubuntu on my Windows computer.
- Install Isaac Lab (particularly need Isaac Gym P4), containing *parallel physics* applications.
- Install unitree\_rl\_gym, which will be utilized to create a training environment for the g1 robot.
- Develop a RL policy for training the g1 to walk

# Logging steps

- Have installed wsl2 previously from previous uses with Ubuntu 22.04.
- nvidia-smi: ensures I have CUDA installed (CUDA 13.1)
- Installed Miniconda3:

```
sudo apt update
sudo apt install wget -y
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
bash Miniconda3-latest-Linux-x86_64.sh
```

- Creating the isaac gym virtual environment (allows us to use specific program versions within the environment):

```
conda create -n g1_rl python=3.8
conda activate g1_rl
python -V      # shows python 3.8.20, the version needed for Isaac Gym environments
```

- Install Pytorch with CUDA within the g1\_rl environment:

```
conda install pytorch torchvision torchaudio pytorch-cuda=12.1 -c pytorch -c nvidia
```

- Within the g1\_rl environment:

code .

-

# Logging steps pt. 2

- Test code to verify Pytorch, CUDA, GPU, and GPU tensors:

```
import torch
print("PyTorch version:", torch.__version__)
print("CUDA available:", torch.cuda.is_available())
if torch.cuda.is_available():
    print("GPU detected:", torch.cuda.get_device_name(0))
    x = torch.rand(3, 3).cuda()
    y = torch.rand(3, 3).cuda()
    z = x + y
    print("GPU tensor addition successful. Result:")
    print(z)
else:
    print("No GPU detected by PyTorch.")
```

# Logging steps pt. 3

- Downloading Isaac Gym Preview 4 (reasoning is it contains actual simulation examples)

- <https://developer.nvidia.com/isaac-gym/download>

- Extracting the archive:

```
tar -xvzf IsaacGym_Preview_4_Package.tar.gz
```

- This allows you to see “isaacgym”

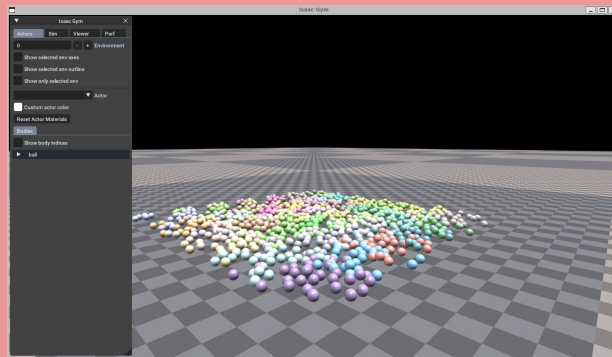
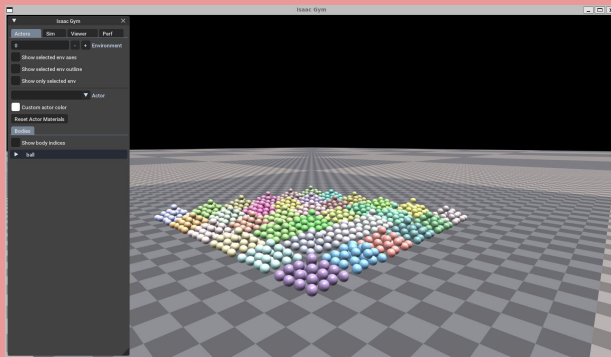
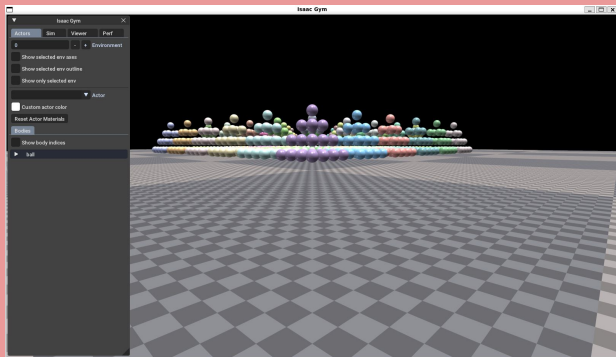
```
cd ~/isaacgym
cd python
pip install -e .
```

- Since I used Ubuntu 22.04, I needed to install python 3.8 dependencies:

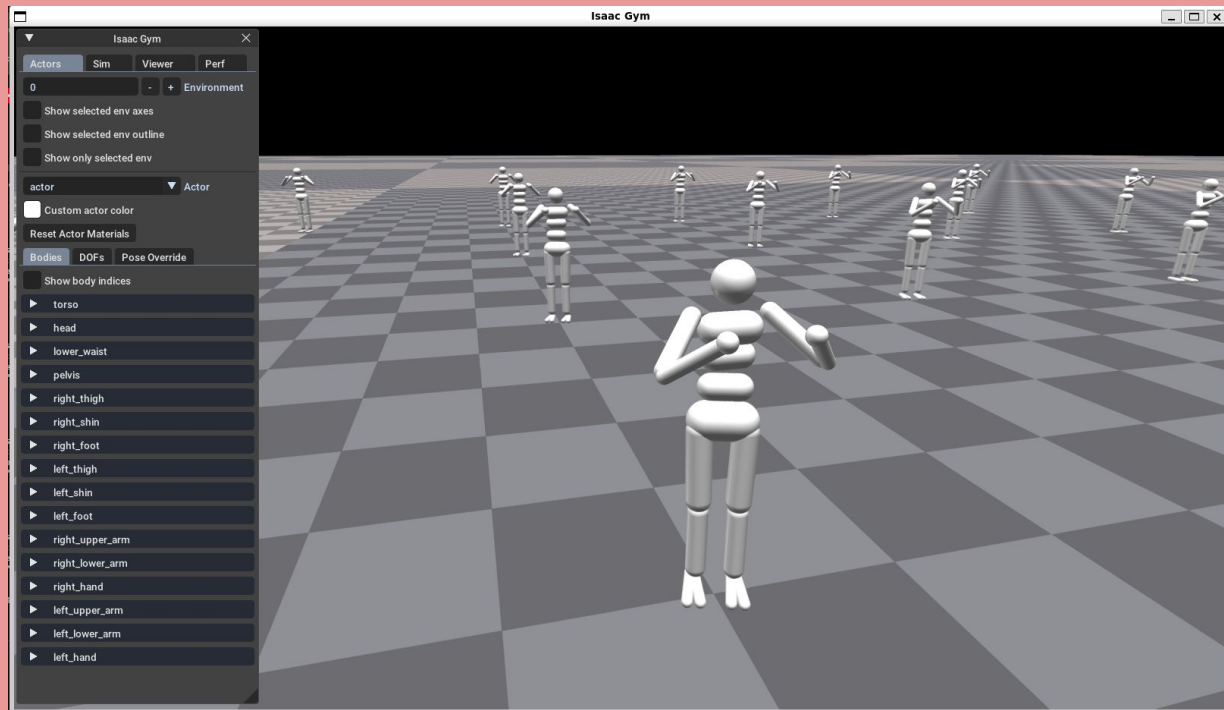
```
sudo apt install libpython3-all-dev
sudo add-apt-repository ppa:deadsnakes/ppa
sudo apt update
sudo apt install python3.8 python3.8-dev
```

- Now in ~/isaacgym/python/examples, we can run python programs that use Isaac Lab.  
Example: `python 1080_balls_of_solitude.py`

# python 1080\_balls\_of\_solitude.py



# python joint\_monkey.py



# Logging steps pt. 4

- Now adding the g1 files (urdf, mesh/.stl, textures) onto the isaacgym assets directory. Putting this in directory “~/isaacgym/assets/g1”
  - [https://github.com/unitreerobotics/unitree\\_ros/blob/master/robots/g1\\_description/g1\\_29dof.urdf](https://github.com/unitreerobotics/unitree_ros/blob/master/robots/g1_description/g1_29dof.urdf)
  - [https://github.com/unitreerobotics/unitree\\_ros/tree/master/robots/g1\\_description/meshes](https://github.com/unitreerobotics/unitree_ros/tree/master/robots/g1_description/meshes)
- Now testing the g1 files on isaacgym.
  - Created a test code that runs the g1 files on isaac gym
  - Created a test code that simulates RL on the g1.

sudo apt install build-essential

sudo apt install ninja-build

- Now creating actual RL code for training g1 robot to walk. First will start with creating a reward policy for standing. Afterwards, will create a reward policy for walking. This way, it should be easier to debug separate issues in the training other than having to debug all of the issues at once.

# Logging steps pt. 5

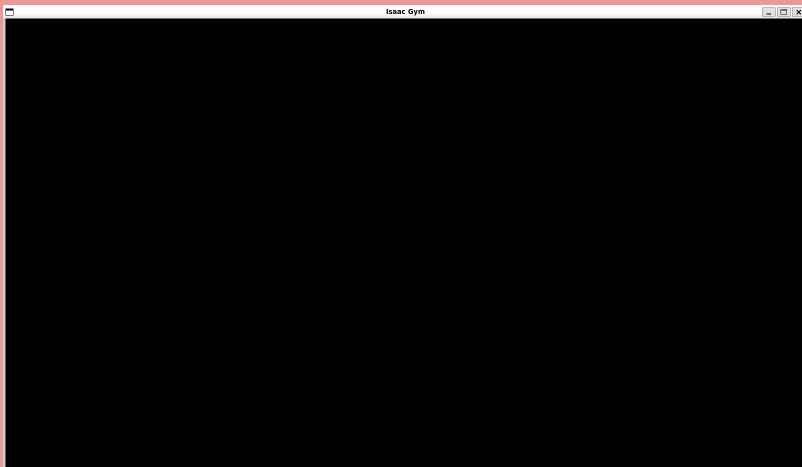
```
git clone https://github.com/leggedrobotics/rsl\_rl.git  
cd rsl_rl/  
git checkout v1.0.2  
pip install -e .
```

```
cd ~  
git clone https://github.com/unitreerobotics/unitree\_rl\_gym.git  
cd unitree_rl_gym/  
pip install -e .  
python legged_gym/scripts/train.py --task g1 #code for opening g1 environment
```

# Current Output

```
[gl_r] mario@LAPTOP-9HUN3CE9:~/untree_rl_gym$ python legged_gym/scripts/train.py --task gl
*** Warning: failed to preload CUDA lib
*** Warning: failed to preload PhysX libs
Importing module 'gym_3d' (/home/mario/isaacgym/python/isaacgym/bindings/linux-x86_64/gym_3d.so)
Setting GymUSU_PLUGIN_PATH to /home/mario/isaacgym/python/isaacgym/bindings/linux-x86_64/used/pluginInfo.json
/home/mario/miniconda3/envs/gl_rl/lib/python3.8/site-packages/torch/_subclasses/functional_tensor.py:258: UserWarning: Failed to initialize NumPy: module compiled against API version 0xf but this version of nu
mPy is 0xe (Triggered internally at /opt/conda/conda-bld/pytorch_1724789259345/work/torch/csrc/utis/tensor_numpy.cpp:84.)
  cpu = _conversion_method_template(device=torch.device("cpu"))
PyTorch version 2.4.1
Device count 1
/home/mario/isaacgym/python/isaacgym/_bindings/src/gymtorch
Using /home/mario/.cache/torch_extensions/py38_cui21 as PyTorch extensions root...
Generating ninja build file /home/mario/.cache/torch_extensions/py38_cui21/gymtorch/build.ninja...
Building extension module gymtorch...
Allowing ninja to set a default number of workers... (overridable by setting the environment variable MAX_JOBS=N)
ninja: no work to do.
Loading extension module gymtorch...
Setting seed: 1
Not connected to PVD
/buildAgent/work/99bede84aa0a52c2/source/physx/src/gpu/PxPhysXGpuModuleLoader.cpp (148) : internal error : libcuda.so!

[Warning] [carb.gym.plugin] Failed to create a PhysX CUDA Context Manager. Falling back to CPU.
Physics Engine: PhysX
Physics Device: cpu
GPU Pipeline: disabled
/home/mario/miniconda3/envs/gl_rl/lib/python3.8/site-packages/torch/functional.py:513: UserWarning: torch.meshgrid: in an upcoming release, it will be required to pass the indexing argument. (Triggered interna
lly at /opt/conda/conda-bld/pytorch_1724789259345/work/aten/src/ATen/native/TensorShape.cpp:3689.)
  return _VF.meshgrid(tensors, **kwargs) # type: ignore[attr-defined]
[Error] [carb.gym.plugin] Gym cuda error: invalid resource handle: ../../source/plugins/carb/gym/impl/Gym/GymPhysX.cpp: 6137
[Error] [carb.gym.plugin] Must enable GPU pipeline to use state tensors
```



# Research

- Isaac-sim shows some locomotion environments for the g1 (in both flat and rough terrain).
- This resource will allow us to learn how to simulate walking with the g1, and therefore we can implement that into simulating the r1.

|                                                                                    |                            |                                                                     |
|------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------|
|  | Isaac-Velocity-Flat-G1-v0  | Track a velocity command on flat terrain with the Unitree G1 robot  |
|  | Isaac-Velocity-Rough-G1-v0 | Track a velocity command on rough terrain with the Unitree G1 robot |

# rough\_env\_cfg.py

- This code is the full-featured G1 locomotion environment in rough terrain.
- The code inherits from *LocomotionVelocityRoughEnvCfg*, providing various simulation parameters (terrain generation, reward manager, event manager, etc.).
  - This is then specialized for the g1

## Steps:

- First, **g1 robot model files** (URDF, meshes) are loaded, and the height scanner location is set for rough terrain perception.
- Next, **randomization events** are implemented
  - Robot joints randomization are set to off (ensures fixed joint configuration)
  - Pose\_range, velocity\_range is set to on
- Then, **reward functions** are implemented (inherited from *RewardsCFG*)
  - Tracking rewards (tracks linear velocity and angular velocity)
  - Gait rewards (tracks feet air time, feet slide penalty)
  - Joint-related penalties (contains penalties for how much a joint can deviate)
  - Termination penalty
  - etc.

# Rough\_env\_cfg.py (continued)

- Next, **Command Ranges** are implemented, ensuring the robot's velocity in any direction is limited, allowing for no turning, sideways walking.
- **Finally, Termination Conditions** are implemented in the case that if the g1's torso falls into the ground; in this case the simulation ends and moves on to the next *episode*.
-

# Rough\_env\_cfg.py (continued)

```
1 # Copyright (c) 2022 2026, The Isaac Lab Project Developers (https://github.com/isaac-sim/isaacsim/blob/main/CONTRIBUTORS.md).
2 # All rights reserved.
3 #
4 # SPDX-License-Identifier: BSD-3-Clause
5
6 from isaacsim.managers import RewardEnvCfg as RewardEnv
7 from isaacsim.managers import SceneEntityCfg
8 from isaacsim.utils import configclass
9
10 import isaacsim.tasks.manager_based.locomotion.velocity.mdp as mdp
11 from isaacsim.tasks.manager_based.locomotion.velocity.velocity_env_cfg import LocomotionVelocityHugbomCfg, RewardsCfg
12
13 #
14 #
15 # Pre-defined configs
16 #
17 from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
18
19 @configclass
20 class G1Rewards(RewardsCfg):
21     """Reward terms for the MDP."""
22
23     termination_penalty = RewardEnv(funct=mdp.is_terminated, weight=-200.0)
24
25     track_lin_vel_xy_exp = RewardEnv(
26         funct=mdp.track_lin_vel_xy_yaw_frame_exp,
27         weight=1.0,
28         params={"command_name": "base_velocity", "std": 0.5},
29     )
30
31     track_ang_vel_z_exp = RewardEnv(
32         funct=mdp.track_ang_vel_z_world_exp, weight=2.0, params={"command_name": "base_velocity", "std": 0.5})
33
34     foot_slip_time = RewardEnv(
35         funct=mdp.foot_slip_time_positive_sliped,
36         weight=-0.25,
37         params={
38             "command_name": "base_velocity",
39             "sensor_cfg": SceneEntityCfg("contact_forces", body_names=["*_ankle_roll_link"],
40             "threshold": 0.4,
41         },
42     )
43
44     foot_slide = RewardEnv(
45         funct=mdp.foot_slide,
46         weight=-0.1,
47         params={
48             "sensor_cfg": SceneEntityCfg("contact_forces", body_names=["*_ankle_roll_link"],
49             "asset_cfg": SceneEntityCfg("robot", body_names=["*_ankle_roll_link"],
50         },
51     )
52
53 # Penalize ankle joint limits
54 def_pos_limits = RewardEnv(
55     funct=mdp.joint_pos_limits,
56     weight=1.0,
57     params={"asset_cfg": SceneEntityCfg("robot", joint_names=["*_ankle_pitch_joint", "*_ankle_roll_joint"])),
58 }
59
60 # Penalize deviation from default of the joints that are not essential for locomotion
61 joint_deviation_hip = RewardEnv(
62     funct=mdp.joint_deviation_hip,
63     weight=-0.1,
64     params={"asset_cfg": SceneEntityCfg("robot", joint_names=["*_hip_yaw_joint", "*_hip_roll_joint"])),
65 }
66
67 joint_deviation_knee = RewardEnv(
68     funct=mdp.joint_deviation_knee,
69     weight=-0.1,
70     params={"asset_cfg": SceneEntityCfg("robot", joint_names=["*_knee_pitch_joint", "*_knee_roll_joint"])),
71 }
72
73 #
74 #
75 # Pre-defined configs
76 #
77 from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
78
79 @configclass
80 class G1LocomotionVelocityHugbomCfg(LocomotionVelocityHugbomCfg):
81     """Locomotion velocity hugbom configuration for the MDP."""
82
83     # Base velocity command
84     base_velocity = CommandCfg(
85         "base_velocity",
86         "std": 0.5,
87     )
88
89     # Ankle roll limits
90     ankle_roll_limits = RewardEnv(
91         funct=mdp.joint_pos_limits,
92         weight=1.0,
93         params={"asset_cfg": SceneEntityCfg("robot", joint_names=["*_ankle_roll_joint"])),
94     )
95
96     # Ankle pitch limits
97     ankle_pitch_limits = RewardEnv(
98         funct=mdp.joint_pos_limits,
99         weight=1.0,
100        params={"asset_cfg": SceneEntityCfg("robot", joint_names=["*_ankle_pitch_joint"])),
101    )
102
103     # Ankle roll deviation
104     ankle_roll_deviation = RewardEnv(
105         funct=mdp.joint_deviation_hip,
106         weight=-0.1,
107         params={"asset_cfg": SceneEntityCfg("robot", joint_names=["*_ankle_roll_joint"])),
108     )
109
110     # Ankle pitch deviation
111     ankle_pitch_deviation = RewardEnv(
112         funct=mdp.joint_deviation_knee,
113         weight=-0.1,
114         params={"asset_cfg": SceneEntityCfg("robot", joint_names=["*_ankle_pitch_joint"])),
115     )
116
117     #
118     #
119     # Pre-defined configs
120     #
121     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
122
123     @configclass
124     class G1SceneEntityCfg(SceneEntityCfg):
125         """Scene entity configuration for the MDP."""
126
127         # Robot
128         robot = SceneEntityCfg(
129             "robot",
130             joint_names=[
131                 "*_shoulder_pitch_joint",
132                 "*_shoulder_roll_joint",
133                 "*_shoulder_yaw_joint",
134                 "*_elbow_pitch_joint",
135                 "*_elbow_roll_joint",
136             ],
137         )
138
139         # Joint deviation flags
140         joint_deviation_flags = RewardEnv(
141             funct=mdp.joint_deviation_hip,
142             weight=-0.1,
143             params={
144                 "asset_cfg": SceneEntityCfg("robot",
145                 joint_names=[
146                     "*_five_joint",
147                     "*_three_joint",
148                     "*_six_joint",
149                     "*_four_joint",
150                     "*_seven_joint",
151                     "*_one_joint",
152                     "*_two_joint",
153                 ],
154             ),
155         )
156
157         # Joint deviation torso
158         joint_deviation_torso = RewardEnv(
159             funct=mdp.joint_deviation_hip,
160             weight=-0.1,
161             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
162         )
163
164     #
165     #
166     # Pre-defined configs
167     #
168     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
169
170     @configclass
171     class G1SceneEntityCfg(SceneEntityCfg):
172         """Scene entity configuration for the MDP."""
173
174         # Robot
175         robot = SceneEntityCfg(
176             "robot",
177             joint_names=[
178                 "*_five_joint",
179                 "*_three_joint",
180                 "*_six_joint",
181                 "*_four_joint",
182                 "*_seven_joint",
183                 "*_one_joint",
184                 "*_two_joint",
185             ],
186         )
187
188         # Joint deviation torso
189         joint_deviation_torso = RewardEnv(
190             funct=mdp.joint_deviation_hip,
191             weight=-0.1,
192             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
193         )
194
195     #
196     #
197     # Pre-defined configs
198     #
199     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
200
201     @configclass
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203         """Scene entity configuration for the MDP."""
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207             "robot",
208             joint_names=[
209                 "*_five_joint",
210                 "*_three_joint",
211                 "*_six_joint",
212                 "*_four_joint",
213                 "*_seven_joint",
214                 "*_one_joint",
215                 "*_two_joint",
216             ],
217         )
218
219         # Joint deviation torso
220         joint_deviation_torso = RewardEnv(
221             funct=mdp.joint_deviation_hip,
222             weight=-0.1,
223             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
224         )
225
226     #
227     #
228     # Pre-defined configs
229     #
230     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
231
232     @configclass
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235
236         # Robot
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238             "robot",
239             joint_names=[
240                 "*_five_joint",
241                 "*_three_joint",
242                 "*_six_joint",
243                 "*_four_joint",
244                 "*_seven_joint",
245                 "*_one_joint",
246                 "*_two_joint",
247             ],
248         )
249
250         # Joint deviation torso
251         joint_deviation_torso = RewardEnv(
252             funct=mdp.joint_deviation_hip,
253             weight=-0.1,
254             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
255         )
256
257     #
258     #
259     # Pre-defined configs
260     #
261     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
262
263     @configclass
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266
267         # Robot
268         robot = SceneEntityCfg(
269             "robot",
270             joint_names=[
271                 "*_five_joint",
272                 "*_three_joint",
273                 "*_six_joint",
274                 "*_four_joint",
275                 "*_seven_joint",
276                 "*_one_joint",
277                 "*_two_joint",
278             ],
279         )
280
281         # Joint deviation torso
282         joint_deviation_torso = RewardEnv(
283             funct=mdp.joint_deviation_hip,
284             weight=-0.1,
285             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
286         )
287
288     #
289     #
290     # Pre-defined configs
291     #
292     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
293
294     @configclass
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300             "robot",
301             joint_names=[
302                 "*_five_joint",
303                 "*_three_joint",
304                 "*_six_joint",
305                 "*_four_joint",
306                 "*_seven_joint",
307                 "*_one_joint",
308                 "*_two_joint",
309             ],
310         )
311
312         # Joint deviation torso
313         joint_deviation_torso = RewardEnv(
314             funct=mdp.joint_deviation_hip,
315             weight=-0.1,
316             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
317         )
318
319     #
320     #
321     # Pre-defined configs
322     #
323     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
324
325     @configclass
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329         # Robot
330         robot = SceneEntityCfg(
331             "robot",
332             joint_names=[
333                 "*_five_joint",
334                 "*_three_joint",
335                 "*_six_joint",
336                 "*_four_joint",
337                 "*_seven_joint",
338                 "*_one_joint",
339                 "*_two_joint",
340             ],
341         )
342
343         # Joint deviation torso
344         joint_deviation_torso = RewardEnv(
345             funct=mdp.joint_deviation_hip,
346             weight=-0.1,
347             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
348         )
349
350     #
351     #
352     # Pre-defined configs
353     #
354     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
355
356     @configclass
357     class G1SceneEntityCfg(SceneEntityCfg):
358         """Scene entity configuration for the MDP."""
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362             "robot",
363             joint_names=[
364                 "*_five_joint",
365                 "*_three_joint",
366                 "*_six_joint",
367                 "*_four_joint",
368                 "*_seven_joint",
369                 "*_one_joint",
370                 "*_two_joint",
371             ],
372         )
373
374         # Joint deviation torso
375         joint_deviation_torso = RewardEnv(
376             funct=mdp.joint_deviation_hip,
377             weight=-0.1,
378             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
379         )
380
381     #
382     #
383     # Pre-defined configs
384     #
385     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
386
387     @configclass
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393             "robot",
394             joint_names=[
395                 "*_five_joint",
396                 "*_three_joint",
397                 "*_six_joint",
398                 "*_four_joint",
399                 "*_seven_joint",
400                 "*_one_joint",
401                 "*_two_joint",
402             ],
403         )
404
405         # Joint deviation torso
406         joint_deviation_torso = RewardEnv(
407             funct=mdp.joint_deviation_hip,
408             weight=-0.1,
409             params={"asset_cfg": SceneEntityCfg("robot", joint_names=["torso_joint"])),
410         )
411
412     #
413     #
414     # Pre-defined configs
415     #
416     from isaacsim.assets import G1_MINIMAL_CFG as isort: skip
417
418     @configclass
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422         # Robot
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424             "robot",
425             joint_names=[
426                 "*_five_joint",
427                 "*_three_joint",
428                 "*_six_joint",
429                 "*_four_joint",
430                 "*_seven_joint",
431                 "*_one_joint",
432                 "*_two_joint",
433             ],
434         )
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1235                 "*_four_joint",
1236                 "*_seven_joint",
1237                 "*_one_joint",
1238                 "*_two_joint",
1239             ],
1240         )
1241
1242         # Joint deviation torso
1243         joint_deviation_torso = RewardEnv(
1244             funct=mdp.joint_deviation_hip,
1
```

# flat\_env\_cfg.py

- This program takes rough\_env\_cfg.py and disables the rough terrain implementation to simulate the g1 walking in a flat surface.

```
Code Blame 34 lines (47 loc) · 1.97 KB ·     
```

```
1 # Copyright (c) 2022-2023, The Isaac Lab Project Developers (https://github.com/isaac-sim/isaac)
2 # All rights reserved.
3 #
4 # SPDX-License-Identifier: BSD-3-Clause
5
6 from isaacsim.managers import SceneEntityCfg
7 from isaacsim.utils import configclass
8
9 from .rough_env_cfg import G1RoughEnvCfg
10
11
12 @configclass
13 class G1FlatEnvCfg(G1RoughEnvCfg):
14     def __post_init__(self):
15         # post init of parent
16         super().__post_init__()
17
18         # change terrain to flat
19         self.scene.terrain.terrain_type = "plane"
20         self.scene.terrain.terrain_generator = None
21
22         # no height scan
23         self.scene.height_sensor = None
24         self.observation_policy.height_scan = None
25
26         # no terrain curriculum
27         self.curriculum.terrain_levels = None
28
29         # Rewards
30         self.rewards.tracked_env_vel_x_exp.weight = 1.0
31         self.rewards.lin_vel_x_z.weight = 0.2
32         self.rewards.action_rate_32.weight = 0.005
33         self.rewards dof_act_32.weight = 1.0e-7
34         self.rewards.feet_air_time.weight = 0.75
35
36         self.rewards.feet_air_time.params["threshold"] = 0.4
37         self.rewards dof_torques_32.weight = 2.0e-6
38         self.rewards dof_torques_32.params["assist_cfg"] = SceneEntityCfg(
39             "robot", joint_names=["*_hip_*", "*_knee_joint"]
40         )
41
42         # Commands
43         self.commands.base_velocity.ranges.lin_vel_x = (0.0, 1.0)
44         self.commands.base_velocity.ranges.lin_vel_y = (-0.5, 0.5)
45         self.commands.base_velocity.ranges.ang_vel_z = (-1.0, 1.0)
46
47
48 class G1FlatEnvCfg_PLA(G1FlatEnvCfg):
49     def __post_init__(self) -> None:
50         # post init of parent
51         super().__post_init__()
52
53         # make a smaller scene for play
54         self.scene.num_envs = 50
55
56         self.scene.env_spacing = 2.5
57         # disable randomization for play
58         self.observation_policy.enable_corruption = False
59
60         # remove random pushing
61         self.events.base_external_force_torque = None
62         self.events.push_robot = None
```

# Isaac Sim and Isaac Lab

Isaac Sim does not work on windows subsystem linux. However, it works in windows, so will be using windows to utilize the Isaac Lab example of g1 reinforcement learning.

- Downloading the IsaacLab github repository on windows  
<https://github.com/isaac-sim/IsaacLab/tree/main/source>
- Downloading Isaac Sim version 5.1.0:  
<https://docs.isaacsim.omniverse.nvidia.com/5.1.0/installation/download.html>
- Python 3.10 is required for Isaac Sim 5.1.0, so will create a virtual environment for this folder.

```
python -m venv IsaacLab_env  
IsaacLab_env/Scripts/activate
```

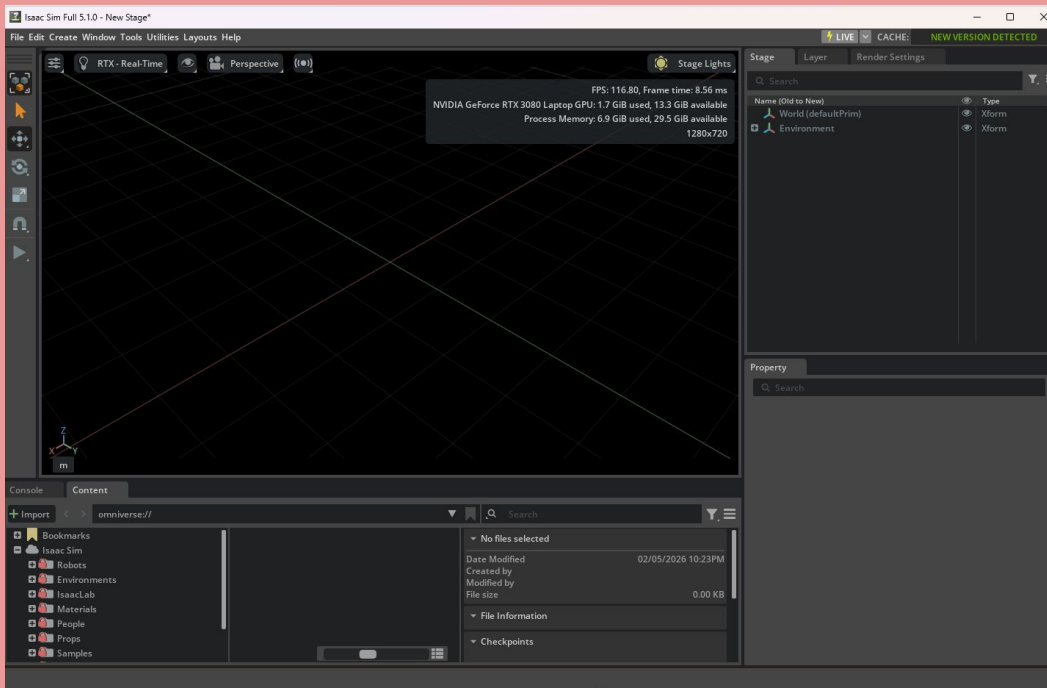
```
cd C:\Users\Mario\IsaacLab
```

```
&
```

```
"C:\Users\Mario\Downloads\isaac-sim-standalone-5.1.0-windows-x86_64\isaac-sim.bat"  
--app "C:\Users\Mario\IsaacLab\apps\isaacclab.app.json" --task G1FlatEnvCfg_PLAY
```

# Isaac Sim and Isaac Lab

```
Windows PowerShell
[19.994s] [ext: omni.kit.converter.hoops-509.1.0] startup
[20.032s] [ext: isaacsim.replicator.grasping-1.0.9] startup
[21.163s] [ext: omni.kit.converter.jt-508.1.0] startup
[21.178s] [ext: omni.services.convert.asset-508.0.2] startup
[21.193s] [ext: omni.kit.quicklayout-1.0.8] startup
[21.199s] [ext: omni.kit.window.commands-0.2.9] startup
[21.207s] [ext: omni.isaac.range_sensor.examples-2.0.6] startup
2026-02-06T03:23:32 [21.043ms] [Warning] [omni.isaac.range_sensor.examples] omni.isaac.range_sensor.examples has been deprecated in favor of isaacsim.sensors.physx.examples. Please update your code accordingly.
2026-02-06T03:23:32 [21.046ms] [Warning] [omni.isaac.range_sensor.examples.lidar_info] omni.isaac.range_sensor.examples.lidar_info has been deprecated in favor of isaacsim.sensors.physx.examples.lidar_info. Please update your code accordingly.
2026-02-06T03:23:32 [21.047ms] [Warning] [omni.isaac.range_sensor.examples.generic_info] omni.isaac.range_sensor.examples.generic_info has been deprecated in favor of isaacsim.sensors.physx.examples.generic_info. Please update your code accordingly.
[21.215s] [ext: isaacsim.util.camera_inspector-1.0.16] startup
[21.239s] [ext: isaacsim.replicator.grasping-ui-1.0.8] startup
[21.247s] [ext: isaacsim.asset.validation-1.1.0] startup
[21.324s] [ext: omni.kit.converter.cad-205.0.0] startup
[21.327s] [ext: omni.kit.window.stats-0.1.7] startup
[21.333s] [ext: isaacsim.app.setup-1.6.0] startup
[21.338s] Isaac Sim Full Version: 5.1.0-rc.19
[21.341s] [ext: omni.kit.profiler.window-2.3.5] startup
[21.367s] [ext: isaacsim.exp.full-5.1.0] startup
2026-02-06T03:23:32 [22.208ms] [Warning] [omni.fabric.plugin] Warning: attribute overrideClipRange not found for bucket id b
[22.561s] app ready
[223.225s] Isaac Sim Full App is loaded.
```



# Sources

<https://docs.robotsfan.com/isaacgym/install.html>

[https://github.com/unitreerobotics/unitree\\_rl\\_gym](https://github.com/unitreerobotics/unitree_rl_gym)

[https://blog.csdn.net/weixin\\_44570248/article/details/146017174](https://blog.csdn.net/weixin_44570248/article/details/146017174)

[https://zsc.github.io/unitree\\_rl\\_gym/doc/setup\\_en.html](https://zsc.github.io/unitree_rl_gym/doc/setup_en.html)

[https://support.unitree.com/home/en/G1\\_developer/rl\\_control\\_routine](https://support.unitree.com/home/en/G1_developer/rl_control_routine)

<https://isaac-sim.github.io/IsaacLab/main/source/overview/environments.html>

<https://gazebo-sim.org/home>